

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

Nov 2012

CE210 MICROPROCESSOR & COMPUTER ARCHITECTURE

Date: 09.11.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions. [11]

- [A] What are the differences between microcontroller and microprocessor? [2]
- [B] Explain terms: Machine Language, Assembly Language, High-level Language [3]
- [C] What is use of following pins of 8085: TRAP, HLDA, INTR [3]
- [D] What is flag? Which are the flags available in 8085 microprocessor? [3]

Q-2 Answer the following questions. [12]

- [A] Write an assembly program to ADD given 5 numbers. [5]
- [B] Explain: Memory Classification [7]

OR

- [B] How does 8085 perform internal data operations? How many registers it has to perform internal data operations? Write functionality of each register. [7]

Q-3 Answer the following questions.(Attempt any Two) [12]

- [A] Draw and explain, connectivity between 8085 and 1K Memory Chip.
- [B] List the sequence of events that occurs when the 8085 CPU reads data from Memory.
- [C] Write down various addressing modes of 8085 with example.
- [D] Draw and explain timing diagram of PUSH instruction.

SECTION-II

Q-4 Answer the following questions. [11]

- [A] How does a microprocessor differentiate between data and instruction code? [2]
- [B] Explain Instruction: CMA, RAR, NOP [3]
- [C] How many address and data lines supported by 8086, 80286, 80386? [3]
- [D] Explain Why latch is used for an output port, but a tri-state buffer can be used for an input port. [3]

Q-5 Answer the following questions. [12]

- [A] What is DMA? How is it interfaced with 8085? [5]
- [B] List the four categories of 8085 instructions that manipulate data. Explain instruction of each category. [7]

OR

- [B] How 8085 does identifies and executes Interrupt service routine? [7]

Q-6 Answer the following questions.(Attempt any Two) [12]

- [A] Draw and explain interfacing between 8255A and 8085 microprocessor.
- [B] Explain the functional diagram of 8155 and explain its control word.
- [C] Write down the differences between memory-mapped I/O and peripheral I/O.